

Genetic Algorithm and Perceptron - Answers

1. Genetic Algorithm

Given: $f(x)=x^2$, Initial Population = {2, 4, 6, 8}.

Step 1: Fitness Calculation

2 → 4
4 → 16
6 → 36
8 → 64

Step 2: Selection

The best individuals are 8 (fitness = 64) and 6 (fitness = 36).

Step 3: Binary Representation

8 = 1000
6 = 0110

Step 4: Crossover

Exchange the least significant bits.

Parent 1 = 1000

Parent 2 = 0110

Since both least significant bits are 0, the offspring remain:

Offspring 1 = 1000 (8)

Offspring 2 = 0110 (6)

Role of Mutation

Mutation randomly changes one or more bits (e.g., 1000 → 1001). It maintains diversity, prevents premature convergence, helps escape local optima, introduces new genetic material, and improves the chance of finding the global optimum.

Final Answer:

Fitness values: 2→4, 4→16, 6→36, 8→64.

Selected individuals: 8 and 6.

Offspring after crossover: 8 and 6.

2. Perceptron Model

Given:

Weights: $w_1 = 0.5$, $w_2 = 0.3$

Bias: $b = -0.4$

Inputs: $x_1 = 2$, $x_2 = 1$

Step 1: Net Input

Net = $w_1x_1 + w_2x_2 + b$
= $(0.5 \times 2) + (0.3 \times 1) - 0.4$
= $1 + 0.3 - 0.4 = 0.9$

Step 2: Step Activation Function

If Net ≥ 0 , output = 1; otherwise output = 0.

Since Net = $0.9 > 0$, Output = 1.

Decision: The customer will purchase the product.

Final Answer:

Net Input = 0.9

Output = 1

Prediction: Customer purchases the product.